BA 875
Operations and Supply Chain Analytics
Homework Assignment 1 (Due 3/25)

Deliverable: Submit your homework assignment as a single Python Notebook only and clearly state your answer to each component of a question. Ensure to save Notebook with output shown. The delivery requirement has been stated here and in the HW submission portal. *You shall receive a 10 point reduction in grade if you do not submit in the format specified*.

Forecasting Price Changes & Promotions
Dataset: jeans_data

You are given the same dataset of prices for a pair of jeans from a medium sized fashion retailer in the southeast USA and their corresponding sales. The data is reported on a week-to-week basis. In addition to price as an input, you are also given a binary indicator variable (se_indicator) specifying whether a given week occurred during a special event (e.g., Christmas).

In the weekly historical data, there was an original retail price for the sales offered ($135) as well as two promotional prices: 15% off ($114.75) and 25% off ($101.25). In a previous analysis (see in-class example), the two 15% off and 25% off promotions that were already tested in stores was examined to see if they had a statistically significant effect. This was done by specifying the two promo prices as dummy variables, as follows:

$$sales = \beta_0 + \beta_1 * promotion114 + \beta_2 * promotion101 + \beta_3 * \text{se_indicator} + \epsilon$$

The above regression is great for retrospectively evaluating the two promotions. But it is not prescriptive. We are interested in giving the retailer a sense of its estimated sales if it decided to offer a brand new promotion to predict sales for a week in the future, i.e., they want you to predict the sales for a future week given you would offer ***promo (P): a "$25 Off" promotion specifically during a special event period***.

As you are interested in predicting future sales from a new type of promotion, you no longer need to examine the previous promotions separately (i.e., no need to include two promo price dummy variables). Rather, you simply need to include 'price' as the variable capturing price differences by week.

Based on this information, reporting the following back to the fashion retailer:

a. Model I: Price (in $) and special event indicator independent variables
   - Calculate the predicted sales from ***promo (P)*** after directly including price and special event indicator as independent variables into a regression model
   - Make sure to interpret your results, including the regression estimation as well as the meaning of your findings based on the prediction for promo (P) - simply showing output of running code does not suffice, provide an interpretation in your own words!

b. Model II: Price (as a % discount off the original price) and special event indicator as independent variables
   - Calculate the predicted sales ***promo (P)*** after directly including price and special event indicator as before, but this time replace the 'price' variable with a % discount variable, i.e. include discount as an independent variable into a regression model special event indicator
   - Make sure to interpret your results, including the regression estimation as well as the meaning of your findings based on the prediction for promo (P) - simply showing output of running code does not suffice, provide an interpretation in your own words!